2406ET099

Roll No. 2

**Course: B. Tech. Computer Engineering/**
**B. Tech. Computer Science and Engineering (AI/ML)**
**Subject: Foundation of Data Analytics,**
**Subject Code: ETCS-110**
**Semester: II**

**Time: 03 Hours** **Max Marks: 70**

**Instructions to the Students:**

1. This Question paper consists of two Sections. All sections are compulsory.
2. Section A comprises 10 questions of short answer type. All questions are compulsory. Each question carries 02 marks.
3. Section B comprises 8 long answer type questions out of which students must attempt any 5. Each question carries 10 marks.
4. Do not write anything on the question paper.

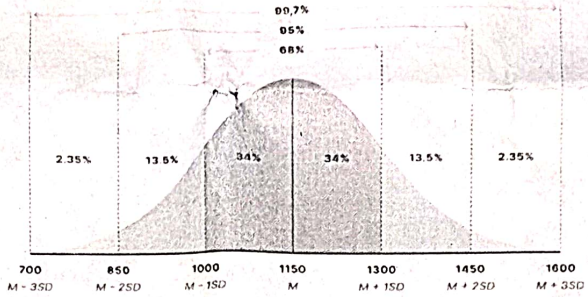
| Q.No. | SECTION –A (SHORT ANSWER TYPE QUESTIONS) | Marks |
|---|---|---|
| 1. a | Explain the following: Discrete Random Variables and Continuous Random Variable. | (2) |
| b | Write the function prototype in python with necessary example. | (2) |
| c | What is the difference between literal and identifier in python | (2) |
| d | Write the python code for Linear searching | (2) |
| e | Explain the continue statement in python using example. | (2) |
| f | What is the difference between LIST and TUPLE. | (2) |
| g | Write python code for Line plot. | (2) |
| h | Write the python code for the sorting the given list using bubble sort: 5, 7, 1, 4, 6, 3, 9. | (2) |
| i | Write the program to add the n even numbers. | (2) |
| j | What is gamma distribution, provide the suitable real-life example where it is followed. | (2) |

**SECTION –B (LONG ANSWER TYPE QUESTIONS)**

| | | |
|---|---|---|
| 2. | What is Poisson Distribution? Explain with the real-life example. How binomia distribution is the approximation to Poisson distribution. | (10) |

3. Write the python code of the following:
   1) To get the input from the user the "NAME" of the person. (2.5)
   2) To print the statement: "What is your name?" (2.5)
   "Hi! I am "NAME". (5)
   3) Write the code for the doubling the given number and print the output.

4. Find which distribution is depictde in the given graph below. Also calculate the probability density function (pdf) of the graph with its derivation and central tendencies. (10)




5. Write a Python function to return sum of first n terms of the following series, where n is taken from user: (10)

$$1+\frac{x^2}{2!}+\frac{x^4}{4!}+\frac{x^6}{6!}+\cdots\ldots\ldots\ldots\frac{x^{2n}}{2n!}$$

6. Create the two 3x3 matrix using NumPy library. Also preform the following operation a) Matrix Addition b) Matrix Multiplication (10)

7. Write a program to create a dictionary of your friends contact number containing name of the friend as the key and Phone number as their values. Compulsory insert 10 entries. (10)

8. Write a python code to write and read in the text file. (10)

9. Explain the function of Matplotlib library. Write the code for design of Histogram of different colours, legend and labels. (10)

===END OF PAPER===